Unit II: Cyber Forensics & Digital Evidence

Digital Forensics Lifecycle

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What is Cyber Forensics?

Cyber Forensics (also called Digital Forensics) is the branch of forensic science that deals with the recovery and investigation of material found in digital devices, preparing evidence that is legally admissible in court.

Evidence Integrity & Law

- **Legal Admissibility:** Forensic investigators must follow strict protocols so that digital evidence is accepted by judges and juries.
- **Chain of Custody:** A chronological paper log documenting who collected, handled, and analyzed the evidence, preventing tampering.
- **Copying Integrity:** Investigators NEVER analyze the original hardware. They extract a bit-stream copy image and analyze the copy.
- **Hash Verification:** Verification that copy matches original using hashing algorithms (MD5, SHA-256).

Forensic Challenges

- **Anti-Forensics:** Techniques used by criminals to hide evidence (encryption, secure file shredding, metadata scrubbing).
- **Data Volume:** Analyzing terabytes of storage on phones and hard drives efficiently.
- **Volatile Memory:** RAM data disappears when the system is powered down, requiring live extraction techniques.
- **Cloud Storage:** Evidence scattered across overseas servers, creating jurisdictional issues.

The Digital Forensics Lifecycle

The forensic process proceeds in five distinct, documented phases:

The Five Forensic Phases

1. Identification: Identify potential evidence sources (computers, phones, USB drives, cloud logs). Document the physical scene.
2. Preservation: Isolate devices from networks to prevent remote wipes (e.g. place phones in Faraday bags). Use hardware write-blockers to copy storage bit-by-bit.
3. Extraction: Recover files, including hidden or deleted data, database records, and browser histories, using forensic software.
4. Analysis: Reconstruct timelines, locate keyword matches, trace emails, and link user actions to the crime. Answer: Who, What, When, and How.
5. Reporting & Presentation: Draft a detailed expert witness report explaining findings clearly for non-technical judges and juries.

Forensic Email Analysis

Email is a primary vector for cybercrime. Forensic analysis involves parsing headers and metadata:

Parsing Email Headers

• **SMTP Hops (Received: fields):** Traces the sequence of mail servers the email passed through, revealing the true sending IP address.
• **Message-ID:** A unique identifier assigned by the sending mail server. Useful for database log auditing.
• **Authentication Checks:** We check:

- **SPF (Sender Policy Framework):** Checks if the sending IP is authorized to mail on behalf of the domain.

- **DKIM (DomainKeys Identified Mail):** Verifies cryptographic signatures to ensure the email was not tampered with in transit.

Tracing Phishing & Spoofing

• **Spoofed Headers:** Attackers modify the 'From:' field display name. However, parsing the raw headers reveals the actual sending email address.
• **Attachment Analysis:** Extracting hidden macro code or malware binaries from PDF/Office attachments inside sandbox test environments.
• **Mail Server Logs:** Auditing SMTP server transaction logs to confirm connections, authentications, and delivery timestamps.

Applications of Cyber Forensics

Where digital forensics is implemented in legal and corporate environments:

⚖️ Criminal Law Case Evidence

Extracts call logs, GPS locations, and deleted chat histories from suspects' smartphones, presenting key timelines to prosecutors.

🏢 Corporate Data Breach Audits

Investigates corporate systems after data breaches. Identifies how hackers entered (the initial entry vector) and logs what data was stolen.

💼 Insider Threat Investigations

Audits logs of employees suspected of leaking intellectual property. Reconstructs file copy actions and USB insert histories.

🔄 Accidental Data Recovery

Assists companies in recovering critical database files deleted by accident or deleted during drive corruptions.

THANK YOU

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